

Capstone Project 2 Report

RAG Pipeline with Multi-AI Agentic Workflow, FastAPI, Docker and CI/CD Deployment

Project Title:

Financial Market Intelligence CrewAI RAG System

Submitted By:

Abhijeet Kumar

Employee ID:

85000067

GitHub Repository:

<https://github.com/Abhijeet-01-bot/financial-market-crewai-rag>



Objective:

The objective of this project is to design and implement a Retrieval-Augmented Generation pipeline enhanced with a multi-agent workflow. The system retrieves relevant financial context from a knowledge base, generates grounded responses using a language model, exposes the workflow through FastAPI, provides a Streamlit frontend, and supports Docker-based deployment with CI/CD automation.

The capstone objective requires a modular RAG pipeline, multi-agent workflow integration, REST API deployment, Streamlit frontend, Docker containerization, GitHub Actions CI/CD, and evaluation using BLEU, ROUGE, and relevance scores.

Problem Statement :

Retrieval-Augmented Generation systems improve trustworthiness by combining language models with external knowledge bases. However, traditional RAG systems often lack structured workflows for planning, reasoning, validation, feedback, and deployment readiness.

This project solves that limitation by implementing an agentic RAG workflow using CrewAI. The system uses specialized agents for retrieval, market analysis, risk assessment, and portfolio allocation.

Project Structure :

financial-market-crewai-rag/

|

├── api/ # FastAPI backend and Pydantic schemas

├── crew/ # CrewAI agents, tasks, tools, and runner

├── data/ # Raw financial text corpus

| └── raw/

├── src/ # RAG, FAISS, evaluation, feedback modules

├── streamlit_app/ # Streamlit chat frontend

├── tests/ # Pytest unit tests

├── vectorstore/ # FAISS vector index

| └── faiss_index/

├── .github/workflows/ # GitHub Actions CI/CD workflow

|

├── Dockerfile # Docker image definition

├── docker-compose.yml # Multi-container deployment

├── requirements.txt # Python dependencies

├── requirements_docker.txt

└── README.md # Project documentation

System Architecture :

User Query

|

v

Streamlit UI

|

v

FastAPI /query Endpoint

|

v

CrewAI Multi-Agent Workflow

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+--> Retriever Agent

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| Financial RAG Search Tool

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v

| FAISS Vector Store

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+--> Market Analysis Agent

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+--> Risk Assessment Agent

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+--> Portfolio Allocation Agent

|

v

Final Financial Intelligence Response

Section 1: RAG Pipeline Development:

Data Preparation :

Financial market data was collected using `yfinance`. The generated text corpus was stored inside: `data/raw/`

The corpus includes :

- `market_reports.txt`
- `stock_news.txt`
- `historical_data.txt`
- `sector_outlook.txt`
- `risk_commentary.txt`

These documents form the knowledge base for the RAG pipeline.

```
AbhijeetKr67@AbhijeetKr67:~$ cd ~/financial-market-crewai-rag
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ ls -lh data/raw
total 20K
-rw-rw-r-- 1 AbhijeetKr67 AbhijeetKr67 2.4K May 11 08:55 historical_data.txt
-rw-rw-r-- 1 AbhijeetKr67 AbhijeetKr67 829 May 11 08:55 market_reports.txt
-rw-rw-r-- 1 AbhijeetKr67 AbhijeetKr67 1.8K May 11 08:55 risk_commentary.txt
-rw-rw-r-- 1 AbhijeetKr67 AbhijeetKr67 1.6K May 11 08:55 sector_outlook.txt
-rw-rw-r-- 1 AbhijeetKr67 AbhijeetKr67 1.2K May 11 08:55 stock_news.txt
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ |
```

>> Financial text corpus generated from web-sourced market data and stored in `data/raw/`.

Embedding and Vector Store :

The financial documents were loaded, preprocessed, split into chunks, embedded, and stored in FAISS.

Embedding model:

sentence-transformers/all-MiniLM-L6-v2

Vector database:

FAISS

FAISS index path:

- vectorstore/faiss_index/

Files generated:

- index.faiss
- Index.pkl

```
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ ls vectorstore/faiss_index/  
index.faiss  index.pkl  
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$
```

- FAISS vector store generated successfully for efficient retrieval.

Retrieval Module:

The retrieval module performs top-k similarity search over FAISS.

File: src/rag_pipeline.py

The retrieval process includes:

1. Load FAISS vector store
2. Embed user query
3. Perform similarity search
4. Return top-k relevant chunks with source metadata

Hybrid retrieval was also implemented by combining semantic FAISS retrieval with keyword-based reranking.

File: src/hybrid_retriever.py

```
(venv) Abhishek@Abhishek007:~/financial-market-crawl-rag$ export PYTHONPATH=$PWD
(venv) Abhishek@Abhishek007:~/financial-market-crawl-rag$ python3 src/rag_pipeline.py
Query:
What is the risk outlook for banking and IT stocks?

Retrieved Context:
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% [ 103/103 [00:00:00, 238.16it/s]

Source 1: sector_outlook.txt
Sector: Banking
Average one-year return of selected stocks: -2.3%
Average annualized volatility: 19.47%
Sector risk level: Medium
Sector outlook: The Banking sector appears neutral or mixed based on recent one-year price performance and volatility of selected listed companies.

Sector: Information Technology
Average one-year return of selected stocks: -28.38%
Average annualized volatility: 23.38%
Sector risk level: Medium
Sector outlook: The Information Technology sector appears weak based on recent one-year price performance and volatility of selected listed companies.

Source 2: risk_commentary.txt
Portfolio Risk Interpretation:
Stocks with higher annualized volatility may increase portfolio drawdown risk. A portfolio concentrated in one sector may face concentration risk. Diversification across banking, IT, FMCG, automobile, and energy sectors can reduce overall exposure.

Risk Appetite Mapping:
Low-risk investors may prefer lower-volatility large-cap and defensive-sector exposure.
Medium-risk investors may combine large-cap stocks, index exposure, and selected sector allocation.
High-risk investors may accept higher volatility and larger equity exposure, but may face larger drawdowns.

Source 3: risk_commentary.txt
ITC.NS | Sector: FMCG | Volatility: 18.98% | Risk Level: Medium | One-year Return: -26.7%
HDFCBANK.NS | Sector: Banking | Volatility: 18.62% | Risk Level: Medium | One-year Return: -28.12%
ICICIBANK.NS | Sector: Banking | Volatility: 17.85% | Risk Level: Low | One-year Return: -11.45%

Source 4: historical_data.txt
Ticker: ICICIBANK.NS
Sector: Banking
One-year starting price: 1437.41
Latest price: 1272.8
One-year return: -11.45%
Annualized volatility: 17.85%
One-year high: 1477.2
One-year low: 1265.9
Calculated risk level based on volatility: Low

Ticker: SBIN.NS
Sector: Banking
One-year starting price: 785.87
Latest price: 979.6
One-year return: 24.63%
Annualized volatility: 21.95%
```

RAG retrieval output showing relevant context from financial knowledge base.

Generation Module:

The generation module uses Google Gemini as the language model. Gemini is called through CrewAI agents to generate structured financial responses conditioned on retrieved context.

Prompt engineering was implemented through:

- CrewAI agent goals
- CrewAI task descriptions
- Expected output instructions
- Financial disclaimer requirements

The output structure includes:

- Market/Sector Trend
- Key Evidence
- Risk Assessment
- Portfolio Allocation
- Disclaimer

Agentic Workflow Development:

CrewAI was used to implement the multi-agent workflow.

Retriever Agent

Retrieves relevant financial context using the Financial RAG Search Tool.

Financial Market Analysis Agent

Analyzes sector outlook, stock performance, market sentiment, and retrieved evidence.

Risk Assessment Agent

Evaluates volatility, downside risk, concentration risk, and sector exposure.

Portfolio Allocation Agent

Suggests educational portfolio allocation based on user risk appetite and previous agent outputs.

Workflow:

Retriever Agent → Analysis Agent → Risk Agent → Portfolio Agent

```
Crew Completion

Crew Execution Completed
Name: crew
ID: 5d51e783-8386-4fa6-8f26-29092b237479
Final Output: Here is an educational portfolio allocation for a user with a medium risk appetite, based on the provided market analysis and risk assessment:

## Educational Portfolio Allocation for Medium Risk Appetite

This allocation is designed for educational purposes and is not financial advice. It aims to balance growth potential with risk mitigation for an investor with a medium risk tolerance.

**Allocation Percentages:**

* **Banking Sector:** 35%
* **Diversified Sectors (FMCG, Automobile, Energy/Conglomerate):** 40%
* **Information Technology (IT) Sector:** 15%
* **Cash/Money Market Instruments:** 10%

**Reasoning and Risk Suitability:**

* **Banking Sector (35%):** The banking sector presents a mixed outlook, offering potential opportunities alongside inherent risks. A significant allocation here acknowledges its foundational role in the economy. For a medium-risk investor, this portion can be further diversified within the sector. For instance, a portion could be allocated to stocks with lower volatility like ICICIBANK.NS, while another part could be allocated to stocks with higher return potential but also higher volatility, such as SBIN.NS, reflecting a balanced approach to risk and reward. The average volatility of 19.47% is moderate, fitting a medium-risk profile.

* **Diversified Sectors (40%):** This is the largest allocation, emphasizing the importance of diversification to mitigate concentration risk.
  * **FMCG (e.g., ITC.NS, HINDUNILVR.NS):** These sectors often provide stability and defensive qualities, even if past returns have been mixed. Their volatility (around 18-20%) is generally manageable.
```

>> CrewAI execution

```
Agent Final Answer

Agent: Financial Market Retriever Agent

Final Answer:
**Banking Sector Outlook:**
The Banking sector exhibits a neutral to mixed outlook based on recent performance. Selected stocks within this sector have shown an average one-year return of -2.3% with an average annualized volatility of 19.47%, classifying it as a medium-risk sector.

**Information Technology Sector Outlook:**
The Information Technology sector presents a weak outlook. Selected stocks in this sector have experienced an average one-year return of -28.38% with an average annualized volatility of 23.38%, also categorized as a medium-risk sector.

**Risk Commentary and Portfolio Suggestions:**
To mitigate portfolio drawdown risk, it is advisable to avoid concentration in a single sector due to potential concentration risk. Diversification across various sectors such as banking, IT, FMCG, automobile, and energy is recommended to reduce overall exposure.

For a medium-risk investor, a suitable portfolio strategy involves a combination of large-cap stocks, index exposure, and strategic allocation to specific sectors. This approach balances potential growth with managed risk.

**Stock Specific Data (Medium Risk):**
* **Banking:**
  * HDFCBANK.NS: Volatility 18.62%, One-year Return -20.12%
  * ICICIBANK.NS: Volatility 17.85%, One-year Return -11.45%
  * SBIN.NS: Volatility 21.95%, One-year Return 24.61%
* **Information Technology:**
  * INFY.NS: Volatility 25.02%, One-year Return -25.39%
  * TCS.NS: Volatility 21.73%, One-year Return -31.4%
* **Other Sectors for Diversification:**
  * ITC.NS (FMCG): Volatility 18.98%, One-year Return -26.7%
```

>> Retriever Agent

```
Agent Final Answer

Agent: Financial Market Analysis Agent

Final Answer:
## Financial Market Intelligence Report

**1. Market/Sector Trend**

* **Banking Sector:** Exhibits a neutral to mixed outlook.
* **Information Technology (IT) Sector:** Presents a weak outlook.

**2. Key Evidence**

* **Banking Sector Performance:** Average one-year return of -2.3% with an average annualized volatility of 19.47%.
  * HDFCBANK.NS: Volatility 18.62%, One-year Return -20.12%
  * ICICIBANK.NS: Volatility 17.85%, One-year Return -11.45%
  * SBIN.NS: Volatility 21.95%, One-year Return 24.61%
* **IT Sector Performance:** Average one-year return of -28.38% with an average annualized volatility of 23.38%.
  * INFY.NS: Volatility 25.02%, One-year Return -25.39%
  * TCS.NS: Volatility 21.73%, One-year Return -31.4%
* **Diversification Evidence:**
  * ITC.NS (FMCG): Volatility 18.98%, One-year Return -26.7%
  * MARUTI.NS (Automobile): Volatility 23.14%, One-year Return 8.55%
  * RELIANCE.NS (Energy / Conglomerate): Volatility 19.74%, One-year Return -2.13%
  * HINDUNILVR.NS (FMCG): Volatility 20.07%, One-year Return -1.4%

**3. Sector or Stock Outlook**

* **Banking Sector:** The mixed performance indicates potential opportunities alongside risks. Stocks like ICICIBANK.NS show lower volatility, while SBIN.NS offers higher return potential but with increased volatility.
* **IT Sector:** The weak outlook and significant negative returns suggest caution. However, for long-term growth potential, a small, strategic allocation might be considered, possibly through diversified IT ETFs or fundamentally strong companies.
```

>> Analysis Agent

```
Agent Final Answer

Agent: Risk Assessment Agent

Final Answer:
1. Risk Level: Medium
2. Volatility Factors: The Information Technology sector exhibits higher average annualized volatility (23.38%) compared to the Banking sector (19.47%). Specific IT stocks like INFY.NS (25.02%) and TCS.NS (21.73%) show significant volatility. Banking stocks also show varying volatility, with SBIN.NS at 21.95% and HDFCBANK.NS at 18.62%. Diversifying stocks like MARUTI.NS (23.14%) and RELIANCE.NS (19.74%) also contribute to overall portfolio volatility.
3. Concentration Risk: High concentration in any single sector, particularly the IT sector with its weak outlook and high volatility, poses a significant concentration risk. Diversification across Banking, IT, FMCG, Automobile, and Energy sectors is recommended to mitigate this.
4. Downside Risks: The IT sector's weak outlook and substantial negative one-year returns (-28.38% average) represent a considerable downside risk. The Banking sector, while mixed, has an average negative one-year return (-2.3%), indicating potential for further declines. Stocks like ITC.NS (-26.7% one-year return) and HINDUNILVR.NS (-1.4% one-year return) also present downside potential.
5. Risk Summary: The overall risk level is assessed as Medium. This is due to the mixed performance of the Banking sector and the weak outlook of the IT sector, both of which have shown negative average returns over the past year. While the IT sector's volatility is higher, both sectors present downside risks. Concentration in any single sector should be avoided. Diversification across Banking, IT, FMCG, Automobile, and Energy sectors is crucial. A balanced approach, potentially including large-cap banking stocks, a cautious allocation to IT, and exposure to other diversifying sectors, is recommended for a medium-risk investor. Specific stock selection within these sectors should consider individual risk-return profiles, such as choosing lower-volatility options like ICICIBANK.NS or higher-return potential with higher volatility like SBIN.NS.
```

>> Risk Agent

```
Agent Final Answer

Agent: Portfolio Allocation Agent

Final Answer:
Here is an educational portfolio allocation for a user with a medium risk appetite, based on the provided market analysis and risk assessment:

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**Allocation Percentages:**

* **Banking Sector:** 35%
* **Diversified Sectors (FMCG, Automobile, Energy/Conglomerate):** 40%
* **Information Technology (IT) Sector:** 15%
* **Cash/Money Market Instruments:** 10%

**Reasoning and Risk Suitability:**

* **Banking Sector (35%):** The banking sector presents a mixed outlook, offering potential opportunities alongside inherent risks. A significant allocation here acknowledges its foundational role in the economy. For a medium-risk investor, this portion can be further diversified within the sector. For instance, a portion could be allocated to stocks with lower volatility like ICICIBANK.NS, while another part could be allocated to stocks with higher return potential but also higher volatility, such as SBIN.NS, reflecting a balanced approach to risk and reward. The average volatility of 19.47% is moderate, fitting a medium-risk profile.

* **Diversified Sectors (40%):** This is the largest allocation, emphasizing the importance of diversification to mitigate concentration risk.
```

>>Portfolio Agent

RAG as Tool Integration:

The RAG pipeline was integrated as a callable CrewAI tool.

File: crew/tools.py

Tool: Financial RAG Search Tool

This allows agents to dynamically query the vector database during reasoning. The retrieved context from the RAG tool is passed to downstream agents, enabling tool chaining and grounded answer generation.

Retrieval Validation:

Retrieval validation was implemented to check whether enough relevant context was retrieved before final answer generation.

File: src/retrieval_validation.py

Validation checks:

- Context exists
- Minimum number of retrieved chunks
- Minimum content length

Evaluation:

The capstone requires measuring performance using BLEU, ROUGE, and relevance scores and comparing baseline RAG with agentic workflow-enhanced RAG.

Evaluation file:

- `src/evaluation.py`

Metrics:

- BLEU
- ROUGE-1
- ROUGE-2
- ROUGE-L
- Retrieval relevance score

Comparison:

Baseline RAG vs Agentic CrewAI RAG

- Evaluation output showing BLEU, ROUGE, and relevance score comparison.

Section 2: REST API and Streamlit Deployment

FastAPI Backend:

The RAG + CrewAI workflow was exposed through FastAPI.

Files: api/app.py

Api/schemas.py

The /query endpoint accepts natural language query and user risk appetite, then returns the generated CrewAI response.

Request body required

Edit Value | Schema

```
{
  "query": "Analyze banking and IT sector outlook for a medium risk investor.",
  "risk appetite": "medium"
}
```

Response body

```
{
  "query": "Analyze banking and IT sector outlook for a medium risk investor.",
  "risk appetite": "medium",
  "response": "Here is an educational portfolio allocation for a user with a medium risk appetite, based on the provided market analysis and risk assessment:\n\n**Educational Portfolio Allocation for a Medium-Risk Investor**\n\nThis allocation is designed to balance potential growth with risk management, considering the current market outlook for the Banking and IT sectors, as well as the general recommendation for diversification.\n\n**Banking Sector:** 35%\n\n**Reasoning:** The banking sector has a neutral to mixed outlook. While the average return is negative, there are individual stocks showing varied performance. ICICIBANK.NS offers a lower risk profile within the sector, while SBIN.NS demonstrates potential for growth despite medium risk. This allocation allows for exposure to potential opportunities while acknowledging the sector's mixed nature.\n\n**Risk Suitability:** Medium. This allocation provides exposure to a sector with moderate volatility and a mixed outlook, aligning with a medium-risk appetite. Careful selection of specific banking stocks within this allocation would be crucial.\n\n**Information Technology (IT) Sector:** 10%\n\n**Reasoning:** The IT sector currently presents a weak outlook with significant negative historical returns. A smaller allocation reflects caution due to its underperformance and higher volatility compared to the banking sector average. This allows for minimal exposure to potential future recovery without over-committing to a currently struggling sector.\n\n**Risk Suitability:** Medium. The IT sector's inherent volatility and recent negative performance make it a higher-risk component. A limited allocation helps to manage this risk for a medium-risk investor.\n\n**Diversified Large-Cap Equities/Index Funds:** 40%\n\n**Reasoning:** To mitigate concentration risk and capture broader market growth, a significant portion is allocated to diversified large-cap equities or index funds. This provides exposure to a wider range of sectors and companies, smoothing out the performance of individual sectors.\n\n**Risk Suitability:** Medium. Large-cap stocks and broad market index funds are generally considered to have moderate risk and volatility, fitting well within a medium-risk profile.\n\n**Consumer Staples/FMCG Sector:** 10%\n\n**Reasoning:** Consumer staples are typically considered defensive and can provide stability to a portfolio, especially during uncertain market conditions. This sector often exhibits lower volatility.\n\n**Risk Suitability:** Low to Medium. This allocation adds a defensive element, helping to cushion potential downturns in more volatile sectors.\n\n**Energy Sector:** 5%\n\n**Reasoning:** The energy sector can offer diversification benefits and potential upside, though it can be cyclical. A smaller allocation provides some exposure without making the portfolio overly sensitive to commodity price fluctuations.\n\n**Risk Suitability:** Medium. The energy sector can experience moderate to high volatility depending on global events and commodity prices.\n\n**Total Allocation:** 100%\n\n**Disclaimer:** This portfolio allocation is for educational purposes only and does not constitute financial advice. It is based on the provided market analysis and risk assessment for a hypothetical medium-risk investor. Investment decisions should always be made after consulting with a qualified financial advisor who can assess your individual financial situation, goals, and risk tolerance. Past performance is not indicative of future results. All investments carry risk, including the potential loss of principal."
}
```

Download

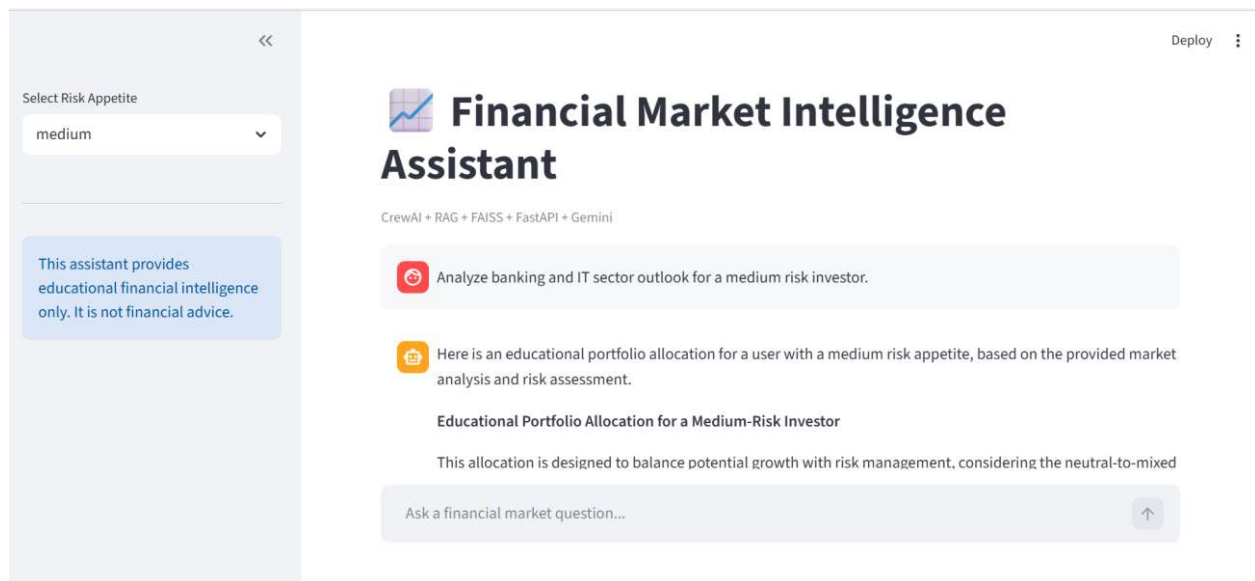
Streamlit Frontend :

A Streamlit chat-style interface was created.

File: streamlit_app/main.py

The frontend allows users to:

- Enter financial queries
- Select risk appetite
- Send query to FastAPI using requests.post()
- Display generated response in chat interface



- Streamlit chat interface for Financial Market Intelligence Assistant.

Section 3: CI/CD Deployment with GitHub Actions and Docker:

A Dockerfile was created to containerize the application.

Docker Compose was used to run both FastAPI and Streamlit.

```
Last login: Mon May 11 19:23:54 2026 from 45.117.203.19
AbhijeetKr67@AbhijeetKr67:~$ cd ~/financial-market-crewai-rag
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ curl http://127.0.0.1:8000/
{"message": "Financial Market Intelligence CrewAI RAG API is running"}AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ docker
docker ps
CONTAINER ID   IMAGE                                NAMES                                COMMAND                                CREATED        STATUS        PORTS
54488229b971   financial-market-crewai-rag_streamlit  "streamlit run strea..."  3 minutes ago  Up 3 minutes  8000/tcp, 0.0.0.0:8501
->8501/tcp, [::]:8501->8501/tcp  financial-crewai-streamlit
7c35d7e288bf   financial-market-crewai-rag_api        "python -m uvicorn a..."  3 minutes ago  Up 3 minutes  0.0.0.0:8000->8000/tcp
, [::]:8000->8000/tcp          financial-crewai-api
AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ |
```

- Docker Container was created

Unit tests were written using Pytest.

Tests validate:

- FastAPI health endpoint
- RAG retrieval output format
- FAISS retrieval availability

```
(venv) AbhijeetKr67@AbhijeetKr67:~/financial-market-crewai-rag$ export PYTHONPATH=$PWD
python3 -m pytest tests/
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /home/AbhijeetKr67/financial-market-crewai-rag
plugins: langsmith-0.8.3, anyio-4.13.0
collected 2 items

tests/test_api.py . [ 50%]
tests/test_retriever.py . [100%]

===== warnings summary =====
venv/lib/python3.12/site-packages/crewai/agent/core.py:352
venv/lib/python3.12/site-packages/crewai/agent/core.py:352
venv/lib/python3.12/site-packages/crewai/agent/core.py:352
venv/lib/python3.12/site-packages/crewai/agent/core.py:352
/home/AbhijeetKr67/financial-market-crewai-rag/venv/lib/python3.12/site-packages/crewai/agent/core.py:352: DeprecationWarning: deprecated
if self.allow_code_execution:

venv/lib/python3.12/site-packages/crewai/agent/core.py:363
venv/lib/python3.12/site-packages/crewai/agent/core.py:363
venv/lib/python3.12/site-packages/crewai/agent/core.py:363
venv/lib/python3.12/site-packages/crewai/agent/core.py:363
/home/AbhijeetKr67/financial-market-crewai-rag/venv/lib/python3.12/site-packages/crewai/agent/core.py:363: DeprecationWarning: deprecated
if self.reasoning and self.planning_config is None:

tests/test_retriever.py::test_retrieve_context
<frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyPacked has no __module__ attribute

tests/test_retriever.py::test_retrieve_context
<frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyObject has no __module__ attribute

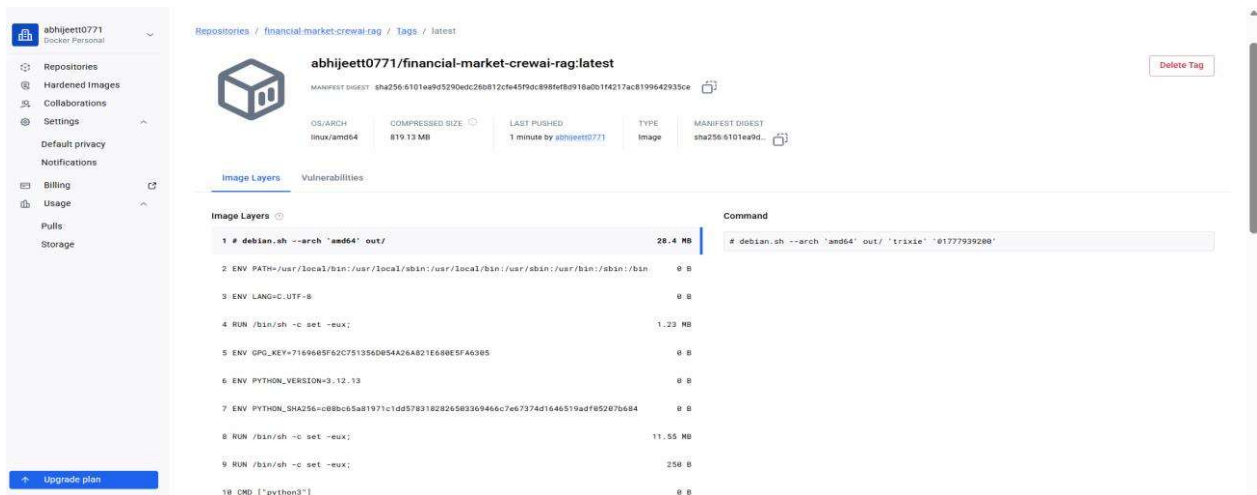
tests/test_retriever.py::test_retrieve_context
<frozen importlib._bootstrap>:488: DeprecationWarning: builtin type swigvarlink has no __module__ attribute

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 2 passed, 11 warnings in 17.66s =====
```

- Pytest output showing all tests passed.

Docker Hub Push:

As part of the CI/CD and deployment requirement, the Docker image for the Financial Market Intelligence CrewAI RAG system was prepared for Docker Hub deployment. Docker Hub allows container images to be tagged and pushed to a remote repository so that they can be reused across different environments. The image was tagged using the Docker Hub namespace and repository name before being pushed to Docker Hub. Docker documentation describes this workflow as tagging the image with namespace/repository:tag and then uploading it using docker push.



Repository: [abhijeet0771/financial-market-crewai-rag](#) / Tags: [latest](#)

abhijeet0771/financial-market-crewai-rag:latest

MANIFEST DIGEST: sha256:6101ea9d5290edc260812cfe45f9dc898f6b918a20114217ac8199642935ce

OS/ARCH: linux/amd64 | COMPRESSED SIZE: 819.13 MB | LAST PUSHED: 1 minute by [abhijeet0771](#) | TYPE: Image | MANIFEST DIGEST: sha256:6101ea9d...

Image Layers

Layer	Command	Size
1	# debian.sh --arch 'amd64' out/	28.4 MB
2	ENV PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin	0 B
3	ENV LANG=C.UTF-8	0 B
4	RUN /bin/sh -c set -eux;	1.23 MB
5	ENV GPG_KEY=7169605F62C7513560B5A2A821E680E5FAA385	0 B
6	ENV PYTHON_VERSION=3.12.13	0 B
7	ENV PYTHON_SHA256=c08bc65a81971c1dd578318282658369466c7e67374d1646519adf652676684	0 B
8	RUN /bin/sh -c set -eux;	11.55 MB
9	RUN /bin/sh -c set -eux;	258 B
10	CMD ["python3"]	0 B

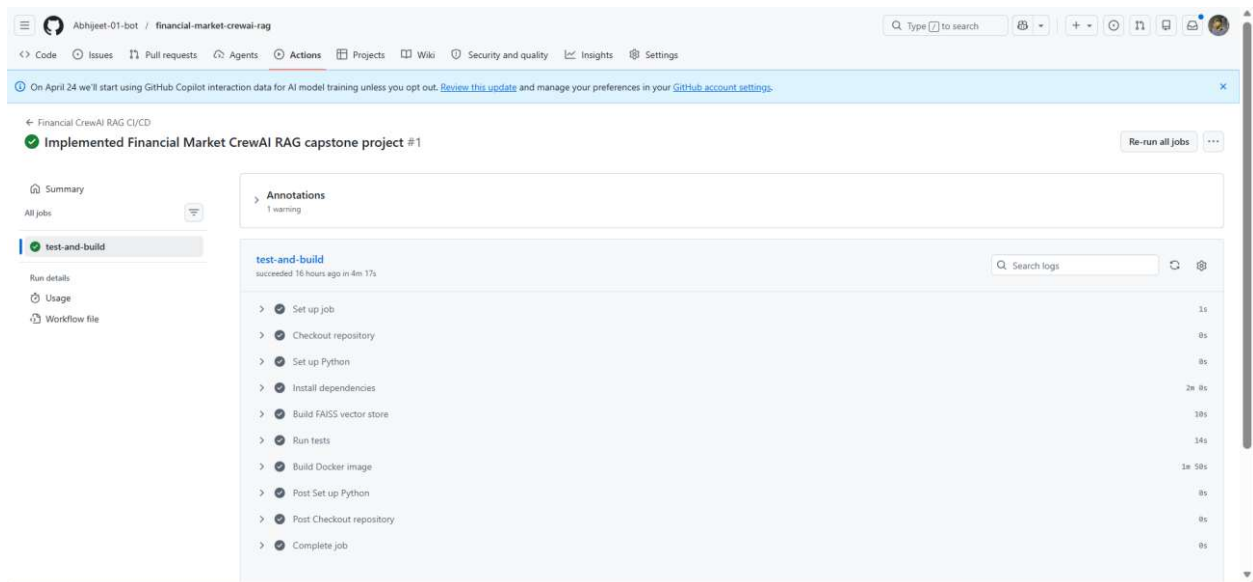
GitHub Actions CI/CD:

A GitHub Actions workflow was created.

File: `github/workflows/ci_cd.yml`

Workflow steps:

- Checkout repository
- Set up Python
- Install dependencies
- Build FAISS vector store
- Run tests
- Build Docker image



- GitHub Actions CI/CD workflow execution

Conclusion:

This capstone project successfully implemented an end-to-end **Retrieval-Augmented Generation pipeline enhanced with a multi-agent workflow**. The system was designed around the Financial Market Intelligence use case, where agents retrieve market data, analyze sector performance, assess risk, and generate educational portfolio allocation suggestions.

The project satisfies the main capstone requirements by implementing a modular RAG pipeline, integrating RAG as a callable tool inside CrewAI agents, exposing the system through FastAPI, providing a Streamlit frontend, containerizing the application with Docker, and validating the workflow through unit testing and CI/CD automation. The capstone document required RAG pipeline development, FastAPI and Streamlit deployment, Docker-based containerization, GitHub Actions CI/CD, evaluation using BLEU/ROUGE/relevance scores, and continuous improvement support. These requirements were addressed through the implemented project modules.